



Investigation of thermal performance at forced convection in plate-fin heat sink

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ABSTRACT

Interest in studies on more effective and faster heat dissipation in microelectronic equipment has increased in recent years due to the cooling area and equipment size limitations. This study attempts to focus on the effect of different fin geometry parameters on pure forced convection heat transfer. Total surface areas of plate-fin heat sinks were kept constant and were carried out for ten different heat sinks in these analyses. The Ansys-Fluent 2023 R1 commercial package program was used to perform the thermal and hydrodynamic performance analyses. Three-dimensional turbulent forced convection heat transfer analyses at heat sinks were subjected to fluid flow ($500 \leq Re \leq 3000$). The effects of gap between fins, fin thickness, fin height, fin number, and air velocity on the heat transfer and the friction factor were obtained and discussed for heat sinks. It was observed that there was a maximum difference of 0.27 % between the maximum temperatures obtained as a result of this analysis and the experimental study results in the literature. Among the ten cases examined (from Case 1 to Case 10), the highest heat transfer coefficient was obtained in Case 2 (number of fins 6, gap between fins 4 mm, fin height 17 mm and fin thickness 2 mm) at $Re = 3000$. The heat transfer coefficient of Case 2 was 41.29 % higher than the heat transfer coefficient of Case 4 at $Re = 3000$; though, the friction factor of Case 2 is 55.93 % higher than Case 4. Also, a useful correlation for the heat transfer coefficient was generated for the fin geometric parameters and Reynolds number.

Nomenclature

A	channel cross-section area, m^2	Nu	Nusselt number
A_t	total heat transfer surface area	p	pressure, Pa
b	base plate thickness, mm	P	wetted perimeter, mm
c_p	specific heat, J/kgK	Q	total heat transfer, W
D_h	hydraulic diameter, mm	Re	Reynolds number
f	friction factor	R_{tc}	thermal resistance, K/W
g	gap between fins, mm	t	fin thickness, mm
h	heat transfer coefficient, W/m^2K	T_∞	air temperature, $^\circ C$
h_{base}	heat transfer coefficient of bare plate, W/m^2K	T_o	air temperature at channel exit, $^\circ C$
h_{eff}	effectiveness of the heat transfer ratio, W/m^2K	T_{base}	base temperature of the heat sink, $^\circ C$
h_{fin}	heat transfer coefficient of plate-fin heat sink, W/m^2K	U_∞	air velocity at channel entrance, m/s
H	height of fin, mm	w	width of heat sink, mm
k	thermal conductivity, W/mK	W	width of channel, mm
L	length of heat sink, mm	α	thermal diffusivity, m^2/s
L_t	total length of the channel	ρ	density, kg/m^3

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(continued)

n	number of fins	ν	kinematic viscosity, m^2/s
		μ	dynamic viscosity, kg/ms

1. Introduction

Developments in electronic technology require effective cooling of electronic equipment. The amount of heat released from electronic equipment such as power converters and thermoelectric generators also directly affects the performance of these equipment. For this reason, the issue of cooling electronic equipment remains current in order to develop faster, more reliable systems. Expanded surfaces, such as heat sinks, and natural convection heat transfer [1–3] or forced convection heat transfer [4–6] methods are commonly used in cooling electronic equipment. Thus, by increasing the heat transfer coefficient (HTC), the amount of heat transfer is maximized. Especially pure forced convection

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is frequently used in heat sink (HS) modules [7]. However, while the amount of heat transfer increases with forced convection, the required fan power must also be taken into account [8]. There are both experimental and numerical studies on different fin structures [9–11], different flow directions [12], different coolants [13], and changes in Reynolds number in forced convection cooling.

Recently, heat transfer in plate fin HSs in laminar forced convection has been investigated in the literature. The heat transfer performance was studied numerically in round, elliptical, and square fin geometries by Soodphakdee vd [14]. They investigated the various fin geometries and examined pressure drop and heat transfer performance. To simplify the analyses, they considered the 2D fully developed laminar forced convection problem. The air velocity ranged from 0.5 to 5 m/s. The results showed that the rounded and staggered geometries were better than sharp-edge fins and inline geometries. Bejan and Morega [15] investigated the optimal array for minimum thermal resistance in fully developed laminar forced convection for round pin fins and staggered parallel-plate fins. Nilpueng et al. [16] experimentally examined the HTC and pressure drop for a plate fin HS with a sinusoidal curve structure and a plate fin HS with a cross-cut sinusoidal curve structure. Nu of sinusoidal wavy plate fin HS was less than that of crosscut sinusoidal wavy plate fin HS by about 5.9–19.1 %. Haque et al. [17] have numerically examined the innovative shaped HS. As a result of the analysis performed for twisted and grooved circular perforated pin fin, they found that the hydrothermal performance factor for grooved circular perforated pin fin increased by 9 %.

It is seen in the literature that heat transfer in different fin geometries such as ring, circular, rectangular, and square have been studied in detail. However, plate fins have become the most preferred fin profiles thanks to their ease of production, low cost, and high efficiency [18]. For this reason, studies examining the effects of rectangular-section fins on heat transfer remain popular. Although studies on heat transfer by both natural and forced convection in plate type fins are available in the literature, these studies are mainly carried out for natural convection [19–24]. When the studies on forced convection heat transfer are examined, it is observed that research has been conducted on thermally and hydraulically developed and developing flows, where fin geometries (fin height, fin spacing, number of fins, etc.) are kept constant.

Teertstra et al. [25] studied forced convection in a plate fin HS and used air as the coolant. They developed an analytical model for high aspect ratio HS. Wu et al. [26] examined the efficiency of plate fin HS. They developed an asymptotic model for hydraulic and thermal performance. When the Nusselt number obtained in the developed asymptotic model was compared with the literature, it was concluded that while the error rate was -15% to $+12\%$ for $Re < 2000$, the error rate was $+7\%$ – 26% for $Re > 2000$. Khudhur D.S. et al. [27] performed the experimental and computational simulation to examine the heat transfer characteristic of two different kinds of straight plate fins at Reynolds number ranging from 6300 to 35120 and heat flux 1194 – 23353 W/m^2 . They concluded that the Nusselt number increased when adding and removing semicircular fins compared to the straight fin structure.

Freegah et al. [28] examined how thermal performance changes in plate fin HSs when the fin base profile was round profile and half-round pins were added to the fins. In addition, the thermal efficiency of half-round pins has been examined for parallel and impinging flow. Culham and Muzchka [29] aimed to minimize entropy production in their study. They optimized the design parameters, airspeed, and fan curve variables of HS to reduce the fluid friction and heat transfer. Iyengar and Bar-Cohen [30] carried out an optimization study to maximize forced convection cooling for a HS covering a certain volume. For this purpose, they presented Coefficient of Performance (COP) analysis for plate fin heat exchangers. They proposed an optimization methodology for minimum material and minimum entropy.

Castelan et al. [31] executed an optimization study to decrease the weight of a HS with plate fins for forced convection. Pujol et al. [32]

studied electrical and thermohydraulic characteristics of a thermoelectric generator (TEG) module and developed an analytical model. Numerical studies were carried out for different fin thicknesses and gaps between fins and the optimum HS design was obtained. They obtained the optimum electrical, thermal and hydraulic behavior for two different HSs (fin thicknesses was 0.3 mm, gap between the fins 1 mm and fin thicknesses was 0.44 mm, gap between the fins 1 mm). In a similar study performed by Elghool et al. [33], TEG performance was investigated experimentally.

According to the detailed literature research, the development of a numerical model that can show the effect of fin geometry parameters on plate fin HSs has not been observed. This study examines the performance of different geometry parameters of fin on pure forced convection heat transfer by keeping the total surface area of the fin constant in a flat plate-type fin arrangement. The focus is on the thermal performances of HSs considered in thermally and hydraulically enhanced flow for different fin thicknesses, fin height, number of fins, gap between fins, and Reynolds numbers. Reynolds number varies between 500 and 3000, and 3D forced convection heat transfer analyses have been performed for the air flow parallel to the fin arrangement. As a result of the analyses, the mean HTC correlation for fin geometry parameters and Re number has been derived and introduced to the literature. To improve the fluid flow and heat transfer behavior of traditional plate-fin HSs, new HSs were designed and developed in this study by changing the fin geometry parameters. Therefore, with the help of this study, it will be possible to develop a HS that requires lower production costs, higher HTC, and lower fan power depending on the fin parameters.

2. Problem definition and mathematical model

This study, the thermal performance of a HS with rectangular plate fins parallel to the flow placed in a channel was discussed. The length (L) of this HS was taken as 68 mm and the width was ($w = n \times t + (n-1) \times g$). Here n was the number of fins, t was the fin thickness, and g was the distance between two fins. The HS was placed $2L$ away from the channel entrance and $0.5L$ away from the channel outlet. The channel width was obtained from the expression ($W = w + 2 \times g$). The only heat source of the system was the heat flux applied to the surface called the base plate

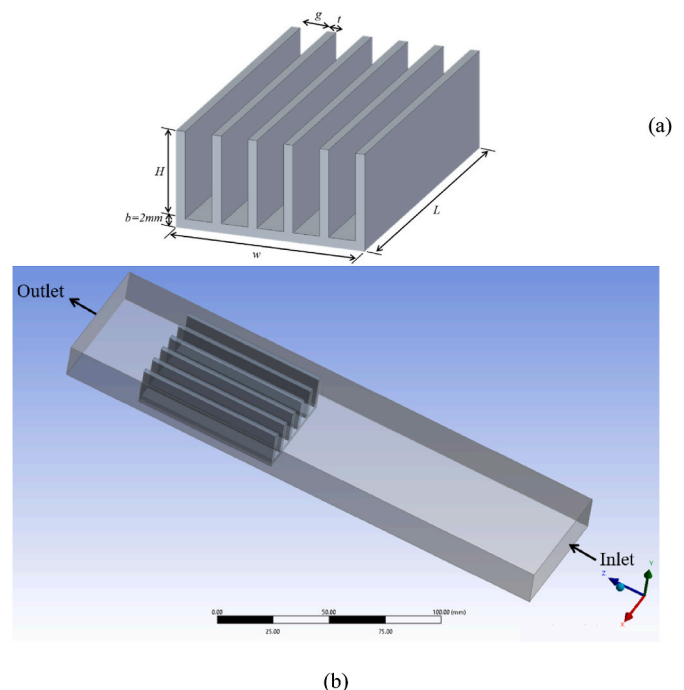


Fig. 1. Rectangular plate fin HS and (b) channel geometry.

($w \times L$) on the HS. The geometric parameters of the HS placed in the channel are given in Fig. 1. The thickness of HS base plate was taken as $b = 2$ mm.

Ten different geometry designs were selected for the rectangular plate finned HS and numerical analyses of these geometries were carried out. The considered geometric parameters are given in Table 1. In the given cases, the heat transfer surface areas were equal to each other ($A_t = 2nLH + g(n-1)L + 2tHn$) [34]. Where H represents the fin height. In order to examine the effects of rectangular plate-finned HS geometry parameters on heat transfer, both the heat transfer area (A_t) and amount of heat transfer from the plate to the system were kept constant.

Case 1 was used for validation in this numerical study. The geometry parameters of Case 1 were the same as the experimental study of Athikari et al. [22]. Other Cases were new heat sink designs obtained with different geometry parameters with the same heat transfer surface area. Thus, a heat sink was determined to provide maximum heat transfer with the same heat transfer surface area.

The governing equations for incompressible flow with constant thermophysical properties in 3-dimensional Cartesian Coordinates are:

Continuity equation:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \quad (1)$$

Momentum equations for x, y, z directions:

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) \quad (2)$$

$$u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial y} + \nu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right) \quad (3)$$

$$u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} = -\frac{1}{\rho} \frac{\partial p}{\partial z} + \nu \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right) \quad (4)$$

Energy equation:

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + w \frac{\partial T}{\partial z} = \alpha \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right) \quad (5)$$

Where u , v and w represent the velocity components in x, y, z directions, ρ density, p pressure, ν kinematic viscosity, T temperature and $\alpha (= k / \rho c_p)$ thermal diffusion coefficient, respectively.

The boundary conditions of the system are given below.

- Channel entrance: The fluid enters the channel at a constant velocity ($U_\infty = \frac{Re \times \nu}{D_h}$) and temperature ($T_\infty = 20^\circ C$).
- Outlet boundary type is defined at the channel exit.
- A heat transfer of $Q = 13.4$ W occurs from the HS base plate to the system. Heat is transferred only from the HS to the fluid. In other words, it is assumed that the channel walls are insulated. Additionally, a no-slip boundary condition was applied on the walls of the HS

and the channel. In numerical analyses, it is assumed that the HS is made of Aluminum 6063 -T5 alloy, and the $k_{Al} = 167 W/mK$.

Air passes through the channel where the HS is located. Thermo-physical properties of air are taken as thermal conductivity $k_{air} = 0.0261 W/mK$, density $\rho = 1.2096 \text{ kg/m}^3$, dynamic viscosity $\mu = 1.915 \times 10^{-5} \text{ kg/ms}$, and specific heat $c_p = 1005 \text{ J/kgK}$ [22]. To examine the flow characteristics, the Re was changed in the range of $500 \leq Re \leq 3000$. The hydraulic diameter calculation ($D_h = 4A/P$), is calculated by using the cross-sectional area of the channel through which the fluid passes ($A = gH$) and the wet perimeter formula ($P = 2H + g$) in this cross-sectional area.

k - kl - ω turbulence model was used in the solution [22]. This model was preferred because it can effectively analyze the transition of the boundary layer from a laminar to a turbulent regime. The model is described by three different equations for turbulent kinetic energy (k_T), laminar kinetic energy (k_L), and inverse turbulent time scale (ω). These equations [35]:

$$\frac{Dk_T}{Dt} = Pk_T + R + R_{NAT} - \omega k_T - Dt + \frac{\partial}{\partial x_j} \left[\left(\nu + \frac{\alpha_T}{\alpha_k} \right) \frac{\partial k_T}{\partial x_j} \right] \quad (6)$$

$$\frac{Dk_L}{Dt} = Pk_L - R - R_{NAT} - D_L + \frac{\partial}{\partial x_j} \left[\nu \frac{\partial k_L}{\partial x_j} \right] \quad (7)$$

$$\begin{aligned} \frac{D\omega}{Dt} = & C_{\omega 1} \frac{\omega}{k_T} Pk_T + \left(\frac{C_{\omega R}}{f_W} - 1 \right) \frac{\omega}{k_T} (R + R_{NAT}) - C_{\omega 2} \omega^2 + C_{\omega 3} f_{\omega} \alpha_T f_W^2 \frac{\sqrt{k_T}}{d^3} \\ & + \frac{\partial}{\partial x_j} \left[\left(\nu + \frac{\alpha_T}{\alpha_\omega} \right) \frac{\partial \omega}{\partial x_j} \right] \end{aligned} \quad (8)$$

The HTC was calculated with Equation (9).

$$h = \frac{\dot{m} c_p (T_o - T_\infty)}{A_t (T_{base} - T_{avr})}, T_{avr} = \frac{(T_o - T_\infty)}{2} \quad (9)$$

Here, $\dot{m}$ is the mass flow rate of the channel, T_o is the channel exit temperature, and T_{base} is the base temperature of the HS.

The following equation is used when calculating the thermal resistance (R_{tc}) [36].

$$R_{tc} = \frac{1}{h A_t} \quad (10)$$

The fin resistance serves to quantify the HS performance as a thermal resistance.

In addition, the friction factor (f) is expressed by the following equation in fully developed internal flows.

$$f = 2 \frac{D_h \Delta p}{\rho L_t U_\infty^2} \quad (11)$$

Here, Δp is the pressure difference at the entrance and exit of the channel, and L_t is the total length of the channel.

3. Numerical solution and verification

Anslys-Fluent 2023 R1 commercial package program was used to perform the analyses. The governing equations were discretized using Finite Volume Method (FVM) and the equations were solved by iterative method using SIMPLE algorithm and Second Order Upwind solver. In turbulent flow solutions, the k - kl - ω turbulence model was used as in Adhikari et al.'s study [22]. A polyhedral grid structure was applied in the study.

Two methods are used to validate computational fluid dynamics (CFD) solutions. The first of these is the method of establishing an experiment set for the current analysis problem, conducting experiments, and comparing the results. The second method is to compare with previous experimental and/or numerical results in the literature. For

Table 1

The considered HS geometry parameters.

Case	n	g (mm)	L (mm)	H (mm)	t (mm)	A_t (m^2)
1	6	6	68	19.0	2	0.0188
2	6	4	68	17.0	6	0.0189
3	6	8	68	16.8	4	0.0189
4	4	10	68	29.0	2	0.0188
5	5	10	68	22.1	2	0.0189
6	4	4	68	27.7	6	0.0188
7	5	4	68	21.3	6	0.0189
8	6	10	68	17.4	2	0.0188
9	6	4	68	19.8	2	0.0188
10	6	8	68	18.2	2	0.0188

numerical verification in this study, the experimental results of Adhikari et al. [22] was used. In the study of Adhikari et al. [18], a HS with a rectangular fin arrangement was discussed. The parameters of the HS are $L=75$ mm, $H=17$ mm, $g=6$ mm, $t=2$ mm, and $n=6$. The total surface area of the fin array is $A_f = 0.01845$ m². The width of the channel is 42 mm and the height of the channel is equal to the fin height and is 17 mm. The heat input from the bottom of the HS is 13.4, 9.67, 7.06, and 4.71 W. In validation analyses, the hydrodynamic inlet length in turbulent flow through the channel was obtained with the formula [36];

$$L_{h,turbulence} = 10 D_h \quad (12)$$

To ensure fully developed flow conditions at the channel entrance, the fins are placed at a distance of $2L$ from the channel entrance.

The results obtained for air velocity values varying between 4 and 10 m/s for $Q = 7.06$ W and $Q = 13.4$ W are presented in Fig. 2. For maximum temperature change, the results obtained from this study, and the results obtained from Adhikari et al.'s [22] experimental studies show similar behavior. While the maximum temperature was same for $Q = 13.4$ W and $U_\infty = 6$ m/s, in $Q = 13.4$ W, the maximum temperature was different for $U_\infty = 4$ m/s, 8 m/s, and 10 m/s. These differences were 0.27 %, 0.1 %, and 0.15 %, respectively. Similarly, for $Q = 7.06$ W and $U_\infty = 10$ m/s, the maximum temperature is almost the same. At $Q = 7.06$ W, the differences between the literature and this study for $U_\infty = 4$ m/s, $U_\infty = 6$ m/s, and $U_\infty = 8$ m/s are 0.16 %, 0.26 %, and 0.23 %, respectively.

Fig. 3 is used for other numerical verification. Fig. 3 shows the changes in the Nusselt number for $Re = 2577.11$, 3865.67, 5154.22, and 6442.78 at $Q = 7.06$ W and $Q = 13.4$ W. At $Q = 13.4$ W, when the Nusselt numbers are compared with the experimental results of Adhikari et al. [18], there is a difference of 4.3 % for $Re = 2577.11$, 0.2976 % for $Re = 3865.67$, 2.0698 % for $Re = 5154.22$ and 3.9503 % for $Re = 6442.78$. On average, the error rate between the experimental results and the analysis was found to be 2.65 %. Similar results were obtained for $Q = 7.06$ W. As a result of the data obtained, it was decided that the model and solution methods were corrected, and analyses were carried out accordingly.

In addition, analyses were carried out for different turbulence models in Ansys-Fluent. In Fig. 4, the maximum temperature obtained from different turbulence models is shown, and the comparison with the experimental results [22] and the correlation results [22] is given. The correlation generated by Adhikari et al. [22] is expressed as $Nu = 0.0019Re + 8.279$. It has been observed that the transition $k-kl-\omega$ turbulence model gives similar results to the experimental results.

The grid independence study was carried out on different mesh structures for Case 1, $Q = 13.4$ W and $Re = 3000$. Within the scope of the analyses, both tetrahedral and polyhedral mesh structures were used

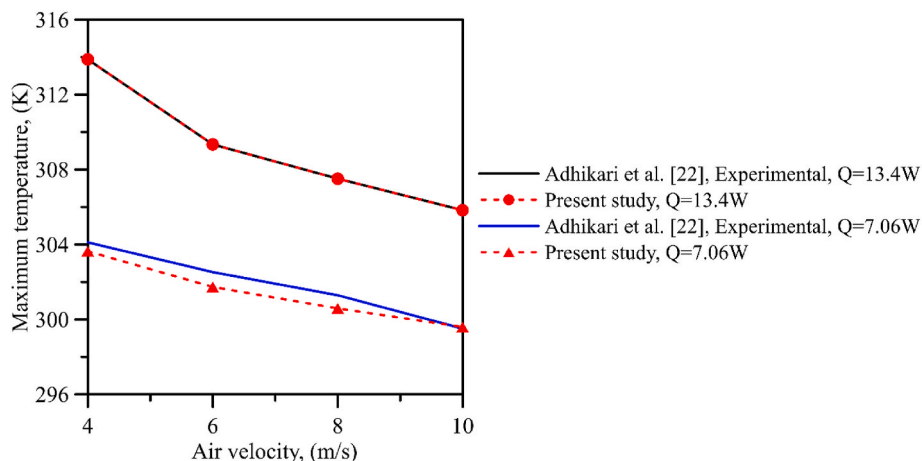


Fig. 2. Comparison of maximum temperature varying with air velocity.

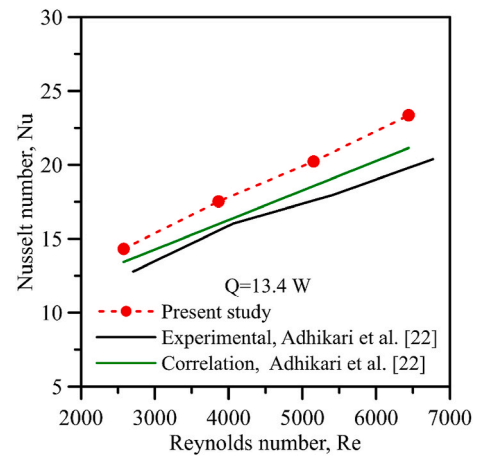


Fig. 3. Comparison of CFD, experimental [22] and correlation [22] results.

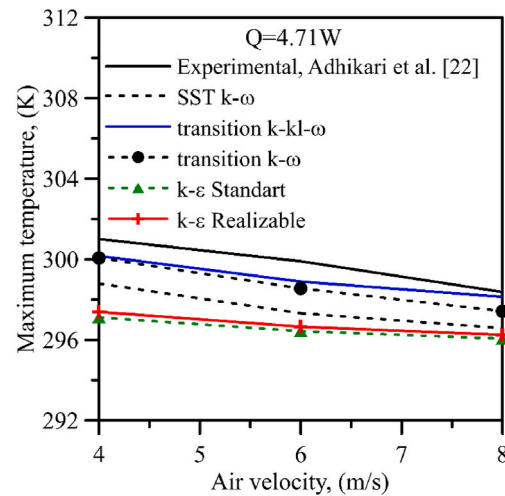


Fig. 4. Comparison of different turbulence models.

and the mean Nusselt number values on the fin surfaces were compared in Fig. 5. According to the analysis results obtained from the mesh independence studies performed according to different mesh structures, the Nusselt number was not effectively changed after the node number 7791822. Therefore, the mesh structure with the number of nodes 9781524 was preferred as the optimum mesh structure. While the

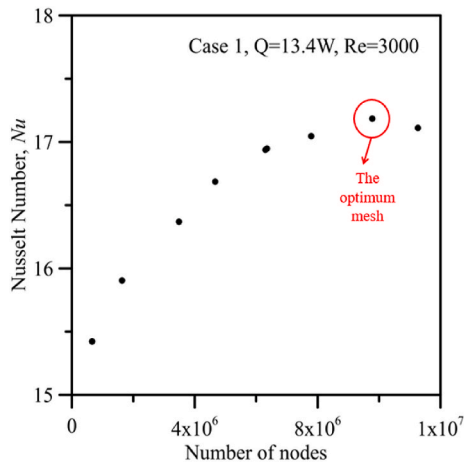


Fig. 5. Grid independence for Case 1, $Q = 13.4\text{W}$, and $Re = 3000$.

Nusselt number change of the selected optimum mesh structure compared to the previous mesh structure (7791822 nodes) is 0.8 %, the difference between the selected optimum mesh structure and the next mesh structure (11263597 nodes) is 0.42 %. While creating the optimum mesh structure, min. size 0.1 mm and max. size was chosen as 0.5 mm and growth rate was chosen as 1.1. In addition, mesh was placed more frequently in 3 layers of all walls. The optimum mesh structure has been applied to all situations [34].

4. Results and discussion

In this study, the plate-fin heat transfer surface areas for each case are equal. Thus, the effects of the gap between fins, fin thickness, and fin height in the fin geometry structure on heat transfer can be examined. To characterize the heat transfer behavior, it is important to examine the HTC and Reynolds number changes. For this reason, the relationship between Re number and HTC in all cases is given in Fig. 6. As is known, as the Re number increases, the HTC increases. In Fig. 6a, the changes in h - Re are examined for the same fin numbers ($n = 6$) and thickness ($t = 2$). While the gap between the fins increases 2.5 times in Case 8 compared to Case 9, the fin height decreases by 0.12 times. The change

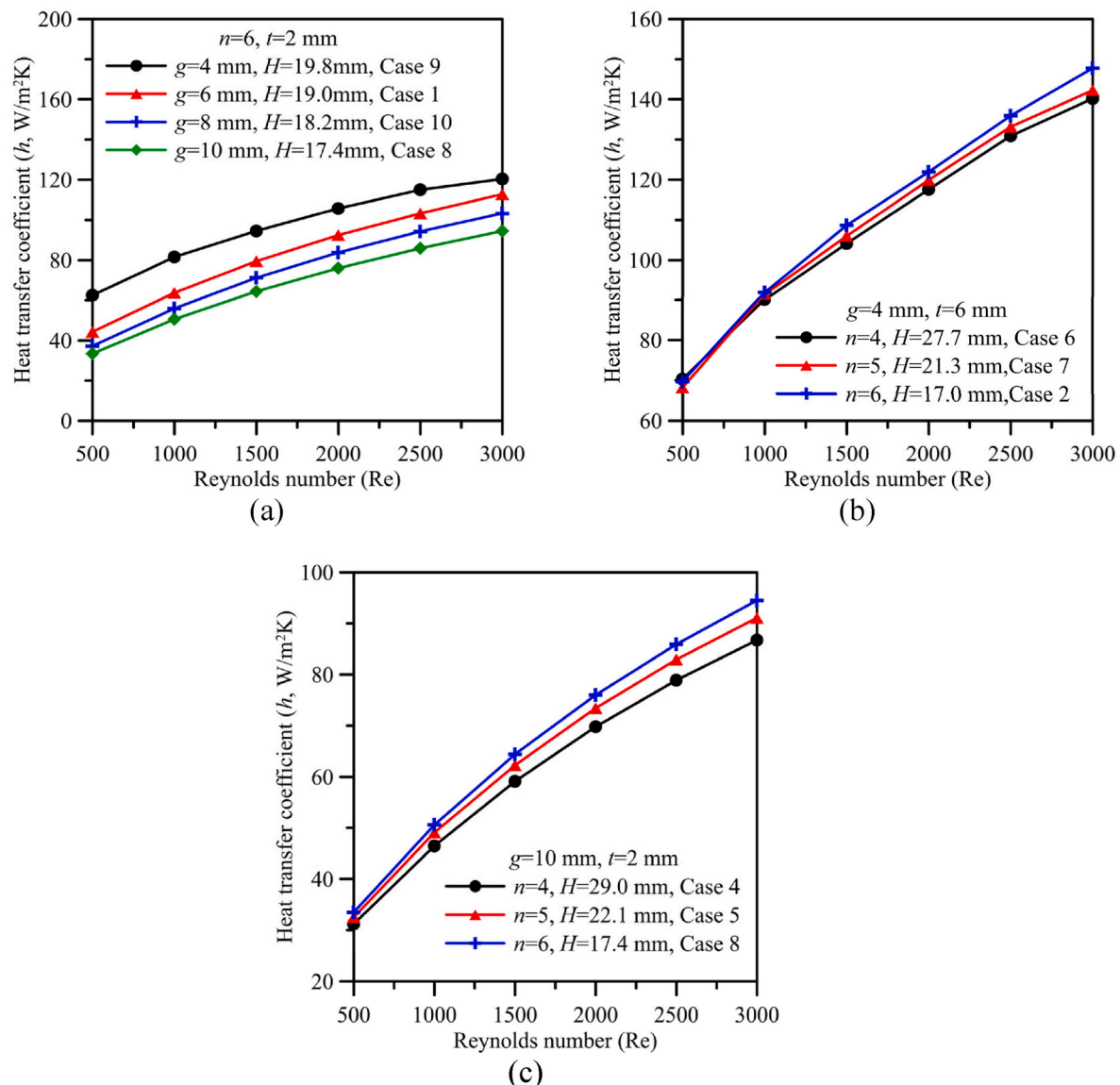


Fig. 6. Variation of HTC according to Re a) $n = 6$ and $t = 2$ mm, b) $g = 4$ mm and $t = 6$ mm, c) $g = 10$ mm and $t = 2$ mm.

in fin height is considerably less than the change in the gap between the fins. Thus, the change in fin height is negligible. Especially as the gap between the fins increases, the HTC decreases significantly. The HTC of Case 9 is an average of 1.5 times higher than that of Case 8 for all Re numbers. In this case, it is concluded that the gap between the fins is an effective parameter of the HTC.

In Fig. 6b, the variation of the HTC with the Re is given for three different cases where the gap between the fins ($g = 4$ mm) and the fin thickness ($t = 2$ mm) are constant and the number of fins and height are variable. It has been observed that the mean HTC increases as the number of fins increases and the fin height decreases. For all Re numbers, the HTC of Case 2 is an average of 3.08 % higher than that of Case 6. The change is small at low Re numbers. A significant difference occurs in the HTC with an increasing Re number while the number of fins and fin height change. While in Case 2, the change in HTC at Re = 500 is 0.84 % in comparison to that of Case 6, this change is 5.37 % for Case 2 compared to Case 6 at Re = 3000. As a result, in cases where the gap between the fins is the narrowest ($g = 4$), the number of fins is higher and the height is lower, causing the HTC to be higher.

In Fig. 6c, the variation of the HTC with the Re is given for three

different cases where the gap between the fins ($g = 10$ mm) and the fin thickness ($t = 6$ mm) are constant, and the number of fins and the fin height change. In the cases where the gap between the fins is the widest ($g = 10$), the change in HTC with the number of fins and fin height created a more significant difference than in Fig. 6b. For all Re numbers, the HTC of Case 4 is 8.65 % higher on average than that of Case 8. As a result, with a wide gap between fins, the number of fins and fin height have a greater effect on the HTC (See Fig. 6b).

Thermal resistance in a HS is defined as a value that indicates how well or poorly it conducts heat. In other words, in systems where a HS is used, the smaller R_{tc} can be considered to be an indicator of effective operation. The variation of R_{tc} depending on the Reynolds number for each HS designed in this study is given in Fig. 7.

For the same number of fins and fin thickness (Fig. 7a), it is seen that the thermal resistance increases as the gap between the fins increases. For Re = 500, the thermal resistance of Case 8 is 1.86 times greater than that of Case 9. As the Re number increases, the difference between the thermal resistances decreases and drops to 1.27 times at Re = 3000. Similar to the change in HTC, it can be concluded that the gap between the fins is also an effective parameter of thermal resistance.

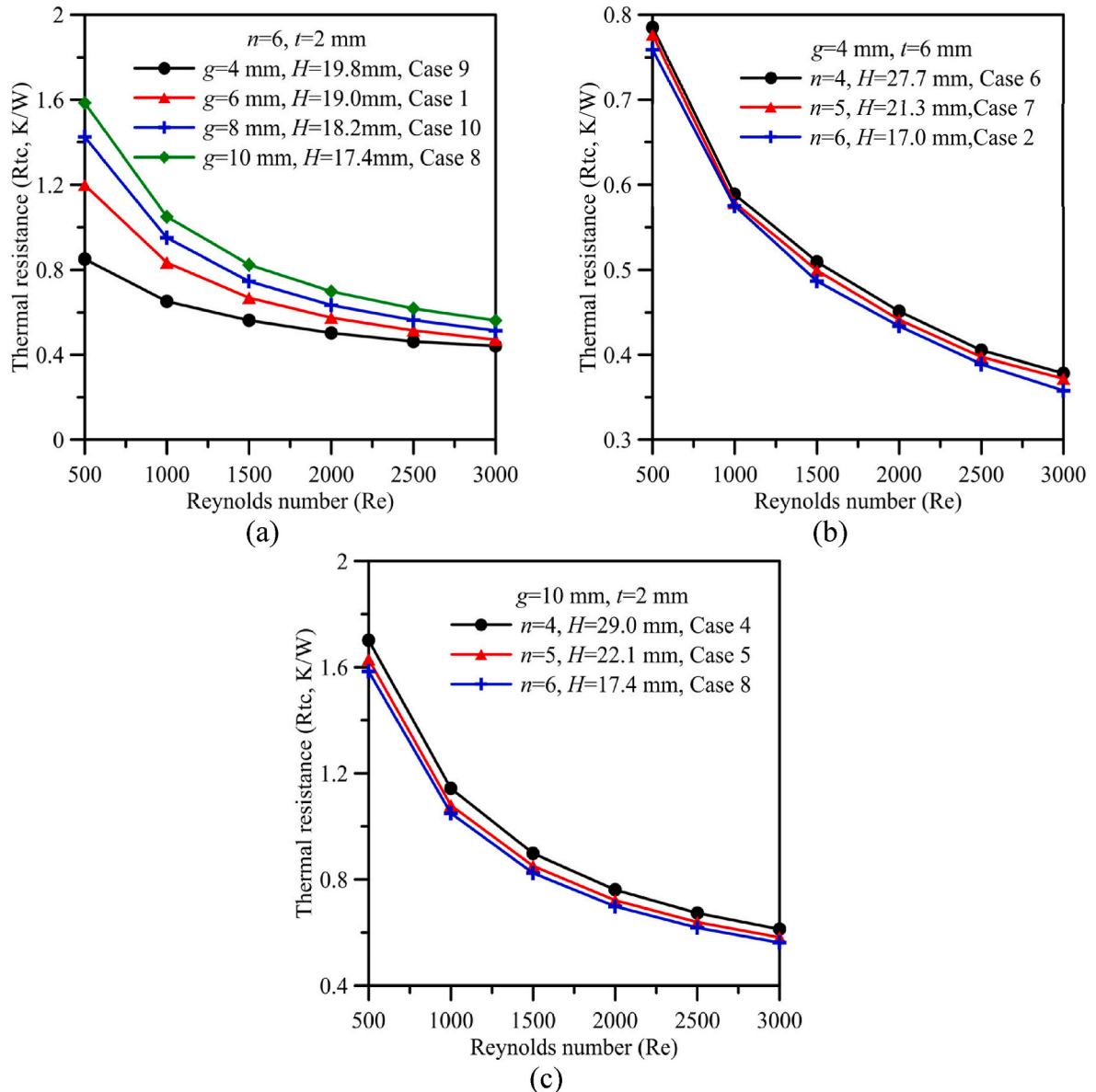


Fig. 7. Variation of thermal resistance according to Re number a) $n = 6$ and $t = 2$ mm, b) $g = 4$ mm and $t = 6$ mm, c) $g = 10$ mm and $t = 2$ mm.

In Fig. 7b, the change in R_{tc} with Re number is given for three different cases where the number of fins and fin height change while the gap between the fins and the fin thickness are kept constant ($g = 4$ and $t = 6$ mm). It has been observed that R_{tc} increases with a decreasing number of fins and an increasing fin height. For all Re numbers, the R_{tc} of Case 2 is on average 4.05 % less than that of Case 6. While at Re = 500 the R_{tc} of Case 2 is less 3.41 % than that of Case 4, this change is 5.69 % at Re = 3000. As a result, in the case where the gap between the fins is the narrowest ($g = 4$), the number of fins is high and the height is low, causing the R_{tc} to be lower.

In Fig. 7c, while $g = 10$ mm and $t = 2$ mm are constant, the number of fins and the fin height change. In this case, the change in thermal resistance with Re number was examined. Similar to Fig. 7b, as the number of fins decreases and the fin height increases, the thermal resistance increases. For all Re numbers, the thermal resistance of Case 4 is an average 8.7 % higher than that of Case 8. As a result, in both cases where the number of fins and fin height change (Fig. 7b and c), R_{tc} decreases as the number of fins increases and the fin height decreases.

The dimensionless pressure drop is expressed by the f . The f is calculated as a function of the pressure gradient in the HS, using the velocity of the fluid into the HS as a reference. The friction factor change for each geometry is given in Fig. 8. As the Re increases, the friction factor decreases. In Fig. 8a, the friction factor value of Case 8 is 1.41 times higher than Case 9.

The variation of the f with Re for cases where the number of fins and

fin height change is given in Fig. 8b. In this graph, the gap between the fins and the fin thickness are constant. It has been observed that as the number of fins decreases and the fin height increases, the friction factor decreases. For all Re numbers, the friction factor of Case 2 is on average 10.19 % higher than Case 6. With increasing Re number and decreasing the number of fins, the friction factor also decreases. The friction factor value of Case 2 is 4.6 % less than Case 6 for Re = 500, while this difference is 10.7 % for Re = 3000. As a result, in the cases where the gap between the fins is narrowest ($g = 4$), the number of fins is high and the height is low, causing the friction factor to be higher.

The variation of the f with the Re is given for three different cases in Fig. 8c. In this Figure, the gap between the fins ($g = 10$ mm) and the fin thickness ($t = 2$ mm) are constant, and the number of fins and fin height change. In the case where the gap between the fins is widest ($g = 10$), the variation of the f with the number of fins and fin height created less difference compared to Fig. 8a and b. In particular, for all Re numbers, the friction factor of Case 8 is higher than Case 4 by an average of 3.87 %. As a result, with a wide gap between fins, the number of fins and fin height have less effect on the friction factor. Although the parameters vary in all three figures, it is generally seen that as the fin height decreases, the friction factor increases.

The HTC, thermal resistance, and friction factor values obtained for all geometries analyzed within the scope of the study are given in Fig. 9. The purpose of presenting the results for all geometries together is to determine the geometry with a high HTC and low thermal resistance and

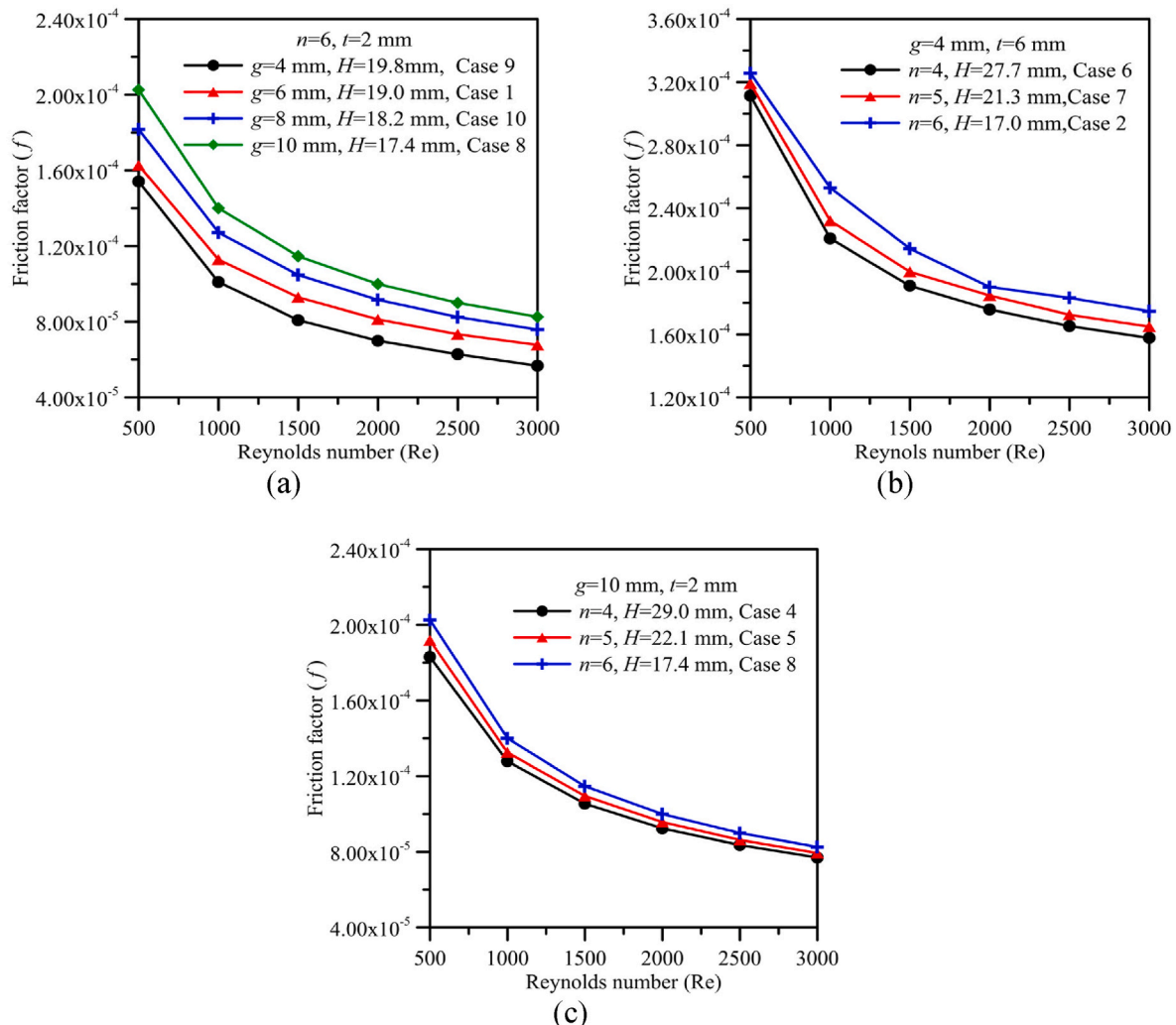


Fig. 8. Variation of friction factor according to Re a) $n = 6$ and $t = 2$ mm, b) $g = 4$ mm and $t = 6$ mm, c) $g = 10$ mm and $t = 2$ mm.

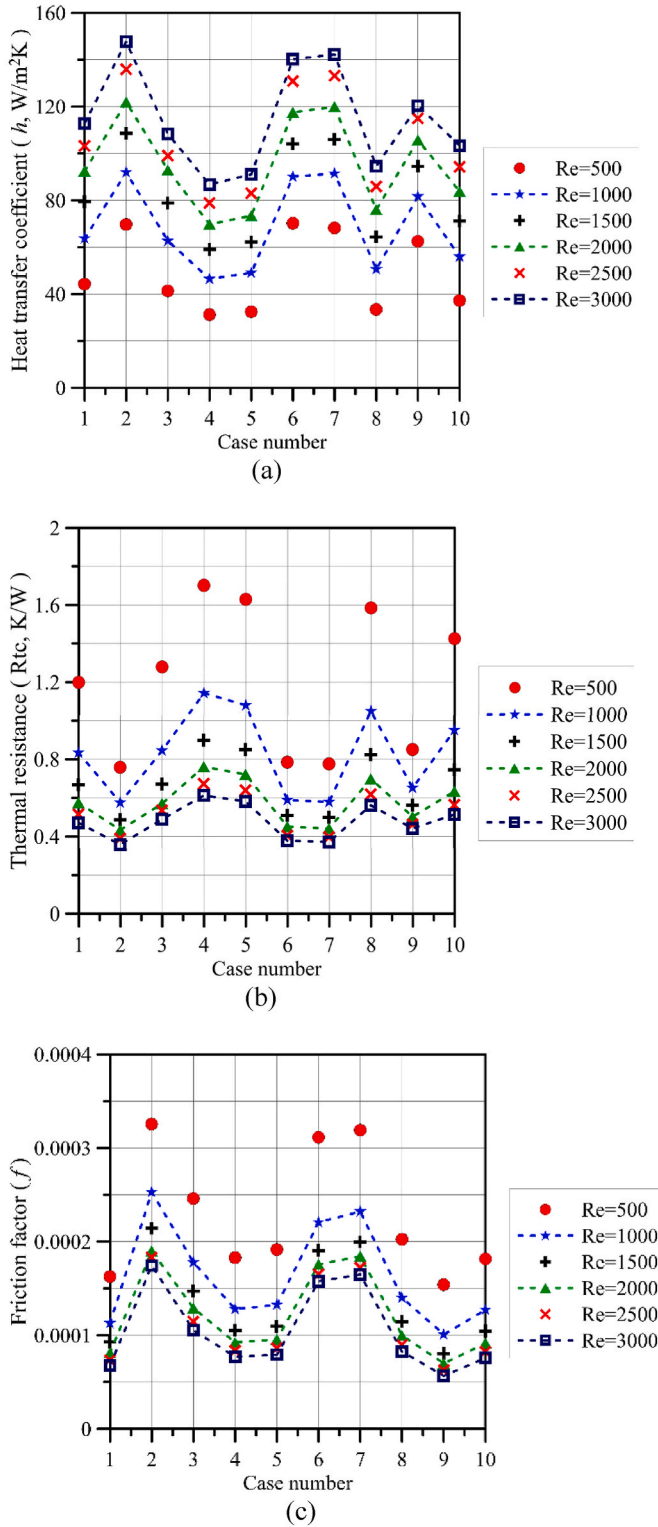


Fig. 9. Comparison of HS cases in terms of (a) HTC, (b) thermal resistance and (c) friction factors.

friction factor. Thus, the optimum geometry with high heat transfer, high thermal resistance and low fan requirement could be determined. The variation of the HTC according to the Re for the solved geometries is given in Fig. 9a. Fig. 9a shows that the highest HTC is obtained for Re = 3000 in Case 2, Case 7, and Case 6, respectively. It is determined that the geometry with the lowest HTC for the same Re is Case 4.

The variation of R_{tc} with Re for each case is shown in Fig. 9b. The

geometry with the highest R_{tc} is Case 4 for Re = 500. The geometry with the lowest R_{tc} for the same Re is Case 2. In the geometry where the R_{tc} is high, the HTC is expected to be small, and in the geometry where the R_{tc} is small, the HTC is expected to be high. As determined in Fig. 9a, the HTC of Case 4 is the lowest and the HTC of Case 2 is the highest, which is compatible with this expectation.

Fig. 9c shows the variation of the f with the Re for the same geometries. The highest friction factor was obtained in Case 2 for Re = 500. The lowest friction factor for the same Re was obtained in Case 9. For Case 1, the difference between the friction factors for Re = 500 and Re = 1000 is 5×10^{-5} , while the difference between the friction factors between Re = 1000 and 3000 is 2.3×10^{-5} .

Fig. 10 shows the changes in heat transfer efficiency at Re = 500 and Re = 3000 for all cases analyzed within the scope of the study. To study the effects of the geometric parameters of the HS on the heat transfer rate, the effectiveness of the heat transfer ratio (h_{eff}) is discussed. The heat transfer efficiency rate (h_{eff}) is defined as the ratio of the HTC (h_{fin}) of the plate-finned HS to the heat transfer coefficient (h_{base}) of the bare plate. To find the h_{eff} ratio, the heat transfer coefficients of the HS in each case for the bare plate case were also analyzed and calculated. This ratio is an indication of the overall effectiveness of the plate-fin HS system design. According to this ratio, it can be known that the system is more efficient in terms of heat transfer when $h_{eff} > 1$. As a result, since h_{eff} in each case is greater than 1, the use of all designed HSs are more efficient than the bare plate. In terms of heat transfer efficiency, it is observed that Case 2, Case 6, and Case 7 are the most efficient designs at both Re = 500 and Re = 3000. Case 4 has the lowest heat transfer efficiency rate.

The temperature distributions of the top views of Case 2 and Case 4 are given in Fig. 11. Here, the design with the highest HTC is Case 2, and the design with the lowest HTC is Case 4. The purpose of considering these two situations is to compare the situations with minimum and maximum heat transfer. Heat input in the system is constant. It was observed that the maximum temperatures in Case 2 were lower than in Case 4. Thus, the heat entering the system was removed better in Case 2. Also, in the temperature distribution obtained for Case 2 and Re = 500, it is seen that the boundary layers developing on the facing surfaces of the adjacent fins overlap. In this case, the temperature difference decreases due to the thick boundary layer on the fin wall. This causes heat transfer to decrease. Such a situation is not observed in Re = 3000. As a result of all analyses, the highest HTC was obtained for Re = 3000.

When the top views of the temperature distribution at Re = 500 and

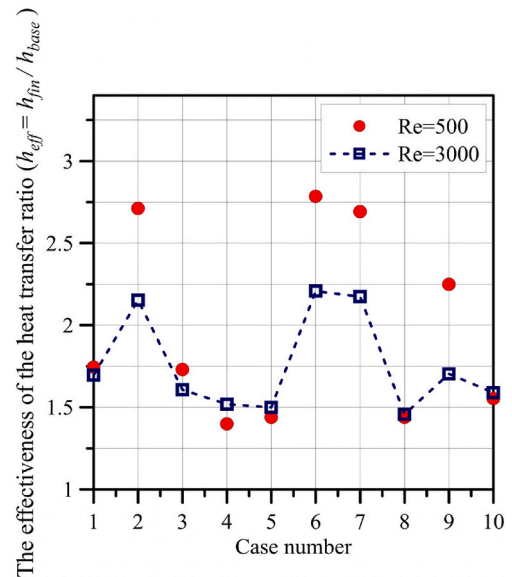


Fig. 10. Change of heat transfer efficiency ratio of all cases for Re = 500 and Re = 3000.

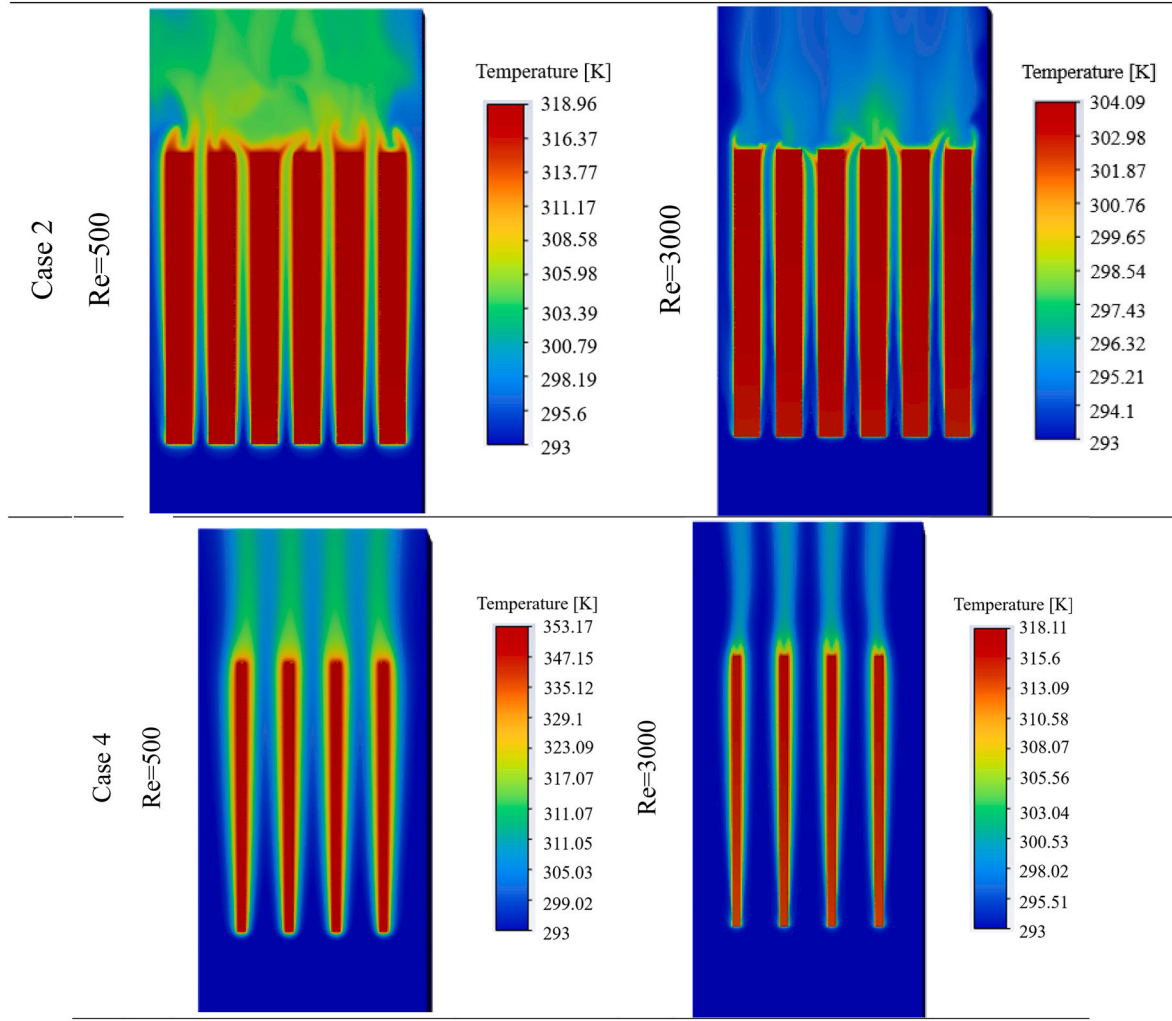


Fig. 11. The temperature distributions of the top view in Case 2 and Case 4.

3000 for Case 4, which has the lowest HTC (Fig. 11), are examined, it is seen that the number of fins and fin thickness are insufficient. In this case, not enough heat was removed from the HS.

Temperature distributions in three different sections in the z -direction in Case 2, where heat transfer convection is the highest, and Case 4, where it is the lowest, are given in Fig. 12. These three cross-sections are close to the entrance ($L_z = 138$ mm), the section in the middle of the HS ($L_z = 170$ mm), and the section of the HS close to the channel outlet ($L_z = 204$ mm). The temperature gradient decreases significantly from the channel entrance to the channel exit, and accordingly, the local heat transfer also decreases significantly. Moreover, this situation is observed clearly at low Re.

Using the Fluent analysis data, the best-fit correlation has been generated. Each of the HS design variables is defined as an independent variable in the correlation. The nonlinear regression for the HTC follows as:

$$h = 0.6881 \frac{\text{Re}^{0.46} \left(\frac{H}{t} \right)^{0.234}}{\left(\frac{g}{w} \right)^{0.894} n^{0.497}} \quad (13)$$

The extremely high r -squared number indicates that the correlation represents 99.7563 % of the data. In Fig. 13, the values obtained from Fluent, and the correlation results derived are compared. As can be seen from the figure, the results are quite consistent. The correlation is valid for the operating range of all design variables. In the figure, only the

comparison of Case 1 and Case 2 is included.

In this study, in order to improve the heat and fluid flow behavior of traditional plate-finned HSs, the HSs were designed and developed by changing the fin geometry parameters. The simulation results have shown that in HS designs where the heat transfer surface area must be kept constant, if the gap between the fins is small, the number of fins must be increased, and the fin height must be reduced. If the gap between the fins is increased, the change in the number of fins and the fin height will significantly affect the HTC.

5. Conclusion

In this study, forced convection heat transfer has been numerically examined in a plate-fin HS with a constant total heat transfer surface area of the fin array when the Re varies between 500 and 3000. Analyses were carried out at varying air velocities, that is, at varying Re, to examine the changes in the Nu . The HS fin array was taken parallel to the flow. To compare the heat transfer performances of finned structures, analyses were also realized for flat plates with finned structure base area. Thus, h_{eff} values were also calculated. As a result of the study, following conclusions can be considered.

- Analyses were carried out for different turbulence models and compared with the experimental results in the literature. As a result of the comparison, it was found that there was an average difference

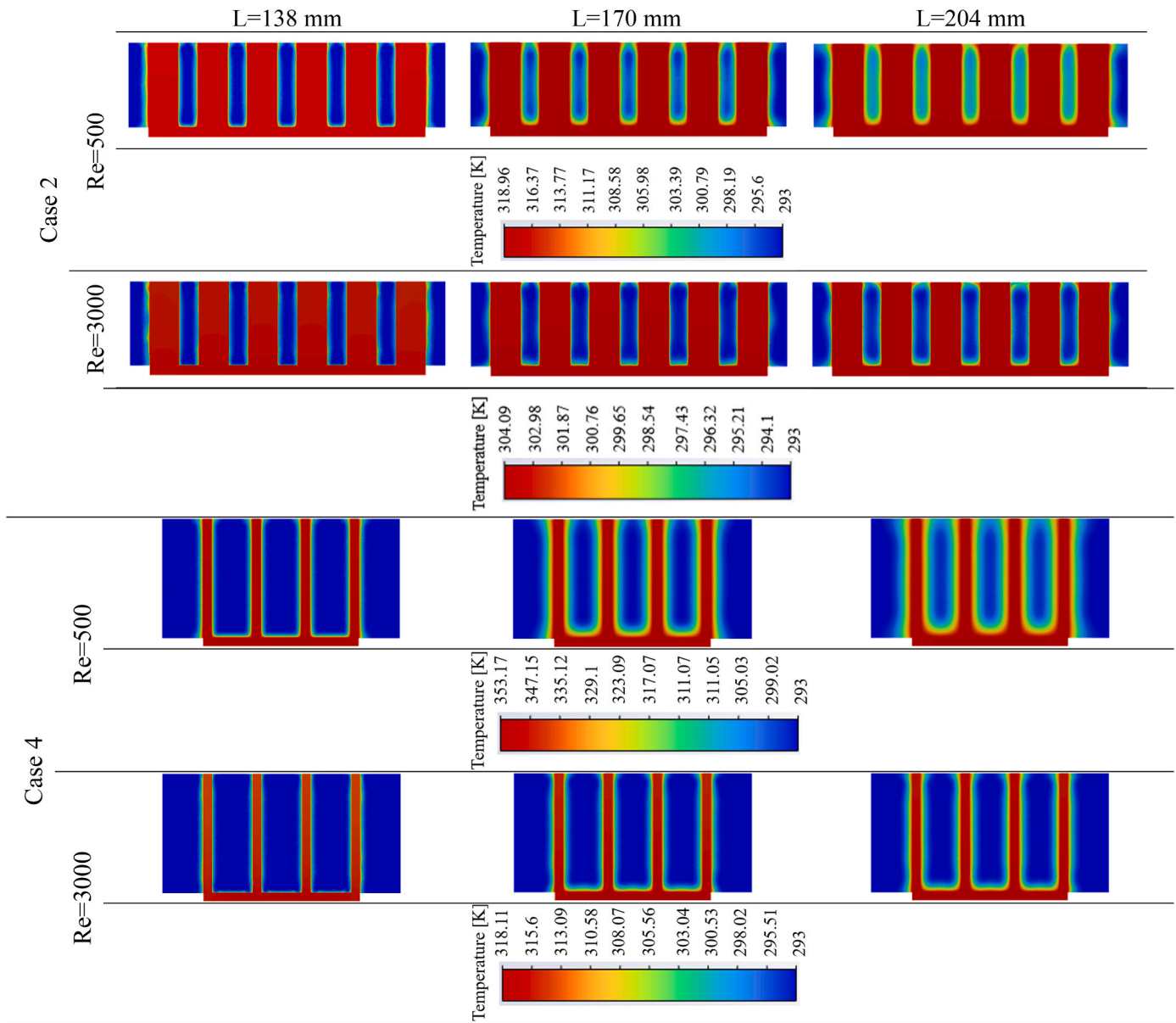


Fig. 12. The temperature distributions in three different cross-sections in the z-direction of Case 2 and Case 4.

of 0.232 % between the results obtained from the $k\text{-}\kappa\text{-}\omega$ turbulence model and the experimental results.

- This study aims to determine the effective design parameter (such as t , g , H , and n) on the HTC of HSs with the same total heat transfer surface area. It has been determined that Case 2 has the highest HTC and is the most effective HS geometry. However, Case 4 was the most inefficient structure on heat transfer. At $Re = 500$ and 3000 , the HTC of Case 2 was 55,21 % and 41,129 % higher than the HTC of Case 4, respectively. In Case 4, due to the small number of fins and the thin thickness of the fins, heat removed from the HS is less than the other Cases.
- Similar to the change in the HTC, it has been concluded that the gap between fins is a more effective parameter on both the HTC and the change in thermal resistance. The HTC of Case 9 is an average of 1.5 times higher than that of Case 8 for all Re numbers.
- If there are situations where the gap between the fins should be small to reduce the R_{tc} , the number of fins should be increased, and the fin height should be reduced.

- For cases with the same number of fins (Case 4 and Case 6, etc.), it has been observed that the fin thickness and the gap between the fins are important in the change of R_{tc} .
- In theoretically, the friction factor used to estimate the required fan power has been calculated based on the fluid velocity and pressure drop. Friction factors in all cases were obtained and compared in this study. While the maximum friction factor value was obtained in Case 2, the minimum friction factor was in Case 9. At $Re = 3000$, the friction factor of Case 2 is about 3 times higher than that of Case 9.
- The correlation of the HTC, r-squared value was 0.997563, was generated for both practical purposes and literature value.

CRediT authorship contribution statement

Ozgun Kosdere: Writing – original draft, Methodology, Conceptualization. **Zerrin Sert:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Formal analysis, Conceptualization. **Ozge Altun:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Formal analysis, Conceptualization.

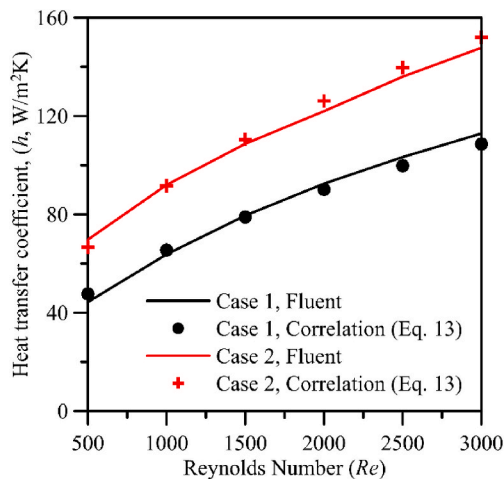


Fig. 13. Comparison of new correlation and CFD analysis for Case 1 and Case 2.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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References

- Shafei H, Abouali O, Jafarpur K, Ahmadi G. Numerical study of heat transfer performance of single-phase heat sinks with micro pin-fin structures. *Appl Therm Eng* 2013;58(1–2):68–76. <https://doi.org/10.1016/j.applthermaleng.2013.04.008>.
- Ahmed HE, Ahmed MI. Optimum thermal design of triangular, trapezoidal and rectangular grooved microchannel heat sinks. *Int Commun Heat Mass Tran* 2015; 66:47–57. <https://doi.org/10.1016/j.icheatmasstransfer.2015.05.009>.
- Peles Y, Kosar A, Mishra C, Kuo CJ, Schneider B. Forced convective heat transfer across a pin fin micro heat sink. *Int J Heat Mass Tran* 2005;48(17):3615–27. <https://doi.org/10.1016/j.ijheatmasstransfer.2005.03.017>.
- Mokheimer EMA. Performance of annular fins with different profiles subject to variable heat transfer coefficient. *Int J Heat Mass Tran* 2002;45:3631–42. [https://doi.org/10.1016/S0017-9310\(02\)00078-9](https://doi.org/10.1016/S0017-9310(02)00078-9).
- Elshafei EAM. Natural convection heat transfer from a heat sink with hollow/perforated circular pin fins. *Energy* 2010;35:2870–7. <https://doi.org/10.1016/j.energy.2010.03.016>.
- Huang GJ, Wong SC, Lin CP. Enhancement of natural convection heat transfer from horizontal rectangular fin arrays with perforations in fin base. *Int J Therm Sci* 2014;84:164–74. <https://doi.org/10.1016/j.ijthermalsci.2014.05.017>.
- Ahmed HE, Salman BH, Kherbeet AS, Ahmed MI. Optimization of thermal design of heat sinks: a review. *Int J Heat Mass Tran* 2018;118:129–53. <https://doi.org/10.1016/j.ijheatmasstransfer.2017.10.099>.
- Hajjalilabaei M, Saghir MZ, Dincer I, Bicer Y. Optimization of heat dissipation in novel design wavy channel heat sinks for better performance. *Energy* 2024;297. <https://doi.org/10.1016/j.energy.2024.131155>.
- Maji A, Bhanja D, Patowari PK. Numerical investigation on heat transfer enhancement of heat sink using perforated pin fins with inline and staggered arrangement. *Appl Therm Eng* 2017;125:596–616. <https://doi.org/10.1016/j.applthermaleng.2017.07.053>.
- He Z, Yan Y, Zhang Z. Thermal management and temperature uniformity enhancement of electronic devices by micro heat sinks: a review. *Energy* 2021;216. <https://doi.org/10.1016/j.energy.2020.119223>.
- Qi D, He J, Xu Y, Lin M, Wang Q. Effect of rib diameter on flow boiling heat transfer with staggered rib arrays in a heat sink. *Energy* 2022;239. <https://doi.org/10.1016/j.energy.2021.122323>.
- Hussain AA, Freegah B, Khalaf BS, Towsyfy H. Numerical investigation of heat transfer enhancement in plate-fin heat sinks: effect of flow direction and fillet profile. *Case Stud Therm Eng* 2019;13:100388. <https://doi.org/10.1016/j.csite.2018.100388>.
- Ye M, Du J, Wang J, Chen L, Varbanov PS, Klemes JJ. Investigation on thermal performance of nanofluids in a microchannel with fan-shaped cavities and oval pin fins. *Energy* 2022;260. <https://doi.org/10.1016/j.energy.2022.125000>.
- Soodphakdee D, Behnia M, Copeland DW. A comparison of fin geometries for heatsinks in laminar forced convection: Part I - round, elliptical, and plate fins in staggered and in-line configurations. *Int J Microcircuits Electron Packag* 2001;24 (1):68–76.
- Bejan A, Morega AM. Optimal arrays of pin fins and plate fins in laminar forced convection. *J Heat Tran* 1993;115(1):75–81. <https://doi.org/10.1115/1.2910672>.
- Nilpuen K, Ahn SH, Jerng D-W, Wongwises S. Heat transfer and flow characteristics of sinusoidal wavy plate fin heat sink with and without crosscut flow control. *Int J Heat Mass Tran* 2019;137:565–72. <https://doi.org/10.1016/j.ijheatmasstransfer.2019.03.114>.
- Haque MR, Hridi TJ, Haque MM. CFD studies on thermal performance augmentation of heat sink using perforated twisted, and grooved pin fins. *Int J Therm Sci* 2022;182:107832. <https://doi.org/10.1016/j.ijthermalsci.2022.107832>.
- Ko YM, Leung CW, Probert SD. Steady-state free-convective cooling of heat exchangers with vertical rectangular fins: effect of fin material. *Appl Energy* 1989; 34(3):181–91. [https://doi.org/10.1016/0306-2619\(89\)90045-7](https://doi.org/10.1016/0306-2619(89)90045-7).
- Jones CD, Smith LF. Optimum arrangement of rectangular fins on horizontal surfaces for free-convection heat transfer. *J Heat Tran* 1970;92(1):6–10. <https://doi.org/10.1115/1.3449648>.
- Dogan M, Sivrioglu M, Yilmaz O. Numerical analysis of natural convection and radiation heat transfer from various shaped thin fin-arrays placed on a horizontal plate-a conjugate analysis. *Energy Convers Manag* 2014;77:78–88. <https://doi.org/10.1016/j.enconman.2013.09.021>.
- Chang S-W, Wu H-W, Guo D-Y, Shi J-j, Chen T-H. Heat transfer enhancement of vertical dimpled fin array in natural convection. *Int J Heat Mass Tran* 2017;106: 781–92. <https://doi.org/10.1016/j.ijheatmasstransfer.2016.09.094>.
- Adhikari R, Wood D, Pahlevani M. An experimental and numerical study of forced convection heat transfer from rectangular fins at low Reynolds numbers. *Int J Heat Mass Tran* 2020;163:120148. <https://doi.org/10.1016/j.ijheatmasstransfer.2020.120418>.
- Iwasaki H, Ishizuka M. Forced convection air cooling characteristics of plate fins for notebook personal computers. In: *Itherm 2000. The seventh intersociety conference on thermal and thermomechanical phenomena in electronic systems* (cat. No. 00CH37069), vol. 2. IEEE; 2000. p. 21–6. <https://doi.org/10.1109/IITHERM.2000.866165>.
- Jouhara H, Axcell BP. Modelling and simulation techniques for forced convection heat transfer in heat sinks with rectangular fins. *Simulation Modelling Practice and Theory* 2009;17(5):871–82. <https://doi.org/10.1016/j.simpat.2009.02.007>.
- Teertstra P, Yovanovich MM, Culham JR. Analytical forced convection modeling of plate fin heat sinks. *J Electron Manuf* 2000;10(4):253–61. <https://doi.org/10.1109/STHERM.1999.762426>.
- Wu H-H, Hsiao Y-Y, Huang H-S, Tang P-H, Chen S-L. A practical plate-fin heat sink model. *Appl Therm Eng* 2011;31:984–92. <https://doi.org/10.1016/j.applthermaleng.2010.10.014>.
- Khudhur DS, Al-Zuhairy ECh, Kassim MS. Thermal analysis of heat transfer with different fin geometry through straight plate-fin heat sinks. *Int J Therm Sci* 2022; 174. <https://doi.org/10.1016/j.ijthermalsci.2021.107443>.
- Freegah B, Hussain AA, Falih AH, Towsyfy H. CFD analysis of heat transfer enhancement in plate-fin heat sinks with fillet profile: investigation of new designs. *Therm Sci Eng Prog* 2020;17. <https://doi.org/10.1016/j.tsep.2019.100458>.
- Culham JR, Muzychka YS. Optimization of plate fin heat sinks using entropy generation minimization. *IEEE Trans Compon Packag Technol* 2001;24(2):159–65. <https://doi.org/10.1109/6144.926378>.
- Iyengar M, Bar-Cohen A. Least-energy optimization of forced convection plate-fin heat sinks. *IEEE Trans Compon Packag Technol* 2003;26(1):62–70. <https://doi.org/10.1109/TCAPT.2003.811484>.
- Castelan A, Cougo A, Dutour S, Meynard T. 3D analytical modelling of plate fin heat sink on forced convection. *Math Comput Simulat* 2019;158:296–307. <https://doi.org/10.1016/j.matcom.2018.09.011>.
- Pujol T, T'Jollyn I, Massaguer E, Massaguer A, C  zar IR, Paape MD. Design optimization of plate-fin heat sink with forced convection for single-module thermoelectric generator. *Appl Therm Eng* 2023;221(25). <https://doi.org/10.1016/j.applthermaleng.2022.119866>.
- Elghool A, Basrawi F, Ibrahim TK, Ibrahim H, M M, Yusof MH, Bagaber SA. Multi-objective optimization to enhance the performance of thermoelectric generator combined with heat pipe-heat sink under forced convection. *Energy* 2020;208. <https://doi.org/10.1016/j.energy.2020.118270>.
- Ko  dere   . Investigation of thermal performance in rectangular fins at low Reynolds numbers [dissertation]. Eskisehir Osmangazi University; 2024.
- Keith Walters D, Coklat Davor. A three-equation eddy-viscosity model for Reynolds-averaged Navier-Stokes simulations of transitional flows. *J Fluid Eng* 2008;130(12). <https://doi.org/10.1115/1.2979230>.
- Incropera FP, DeWitt DP, Bergman TL, Lavine AS. *Fundamentals of Heat and mass transfer*. sixth ed. John Wiley & Sons; 2006.